

Churn-HPC: Executive Technical Report

DNN churn classification across the modern AI infrastructure stack

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1. Problem and Objective

Customer churn prediction on `Churn_Dataset.csv`: 5,000 telecom customers, 16 numeric usage features, boolean `churned` label with a 14% positive rate. The modeling task (a multilayer perceptron reaching 92-94% test accuracy against an 86% majority baseline) is deliberately modest; the engineering objective is the point: run one workload through an immutable, reproducible infrastructure pipeline spanning containerization, orchestration, storage staging, and multi-hardware benchmarking, and identify the true performance bottleneck with measured telemetry rather than intuition.

The resource challenge is the inverse of the usual one. The dataset is 341 KB; every layer of the stack (10-core CPU node, Apple M4 GPU, provisioned NVIDIA L4 and TPU v5e targets) is oversized for it. That makes it a clean probe of **fixed costs**: compilation, kernel dispatch, staging, orchestration overhead. These are exactly the costs that silently dominate small-to-medium scientific workloads on shared clusters.

2. Architecture

Phase 1 - Immutable environment. Two Dockerfiles (`docker/`): a CPU image on `python:3.12-slim` and a CUDA 12.4 + cuDNN image carrying the pinned `jax[cuda12]` /XLA and PyTorch stack. All dependencies are baked at build time from pinned requirements files; running nodes install nothing. On SLURM the same images run read-only via Apptainer (`.sif`).

Phase 1 - Orchestration manifests. Kubernetes Jobs (`k8s/`) declare exact hardware constraints: the GPU job requests and limits 8 CPU cores, 24 GiB RAM, and exactly 1 `nvidia.com/gpu` on an L4 node pool; the TPU job pins a v5e 2x2 slice (4 chips, 24 cores, 48 GiB). SLURM batch scripts (`slurm/`) mirror this with `--cpus-per-task`, `--mem`, and `--gres=gpu:1`, plus a 2-node `jax.distributed` variant with the coordinator pinned to the first node of the allocation.

Phase 2 - Provisioning and storage. Three mapped targets: the local M4 node (measured), the Stanford ME344 SLURM cluster `hpcc-cluster-49` (VPN-gated, manifests ready), and a private VPC-native GKE cluster (subnet `10.10.0.0/20`, pod/service secondary ranges, no public node IPs, egress via Cloud NAT). The data tier funnels the CSV as: shared NFS or GCS canonical copy, staged once per job to node-local NVMe scratch (`$SLURM_TMPDIR / local-SSD emptyDir / gcsfuse file cache`), then loaded to RAM once and pushed to device memory. The training loop never touches shared storage.

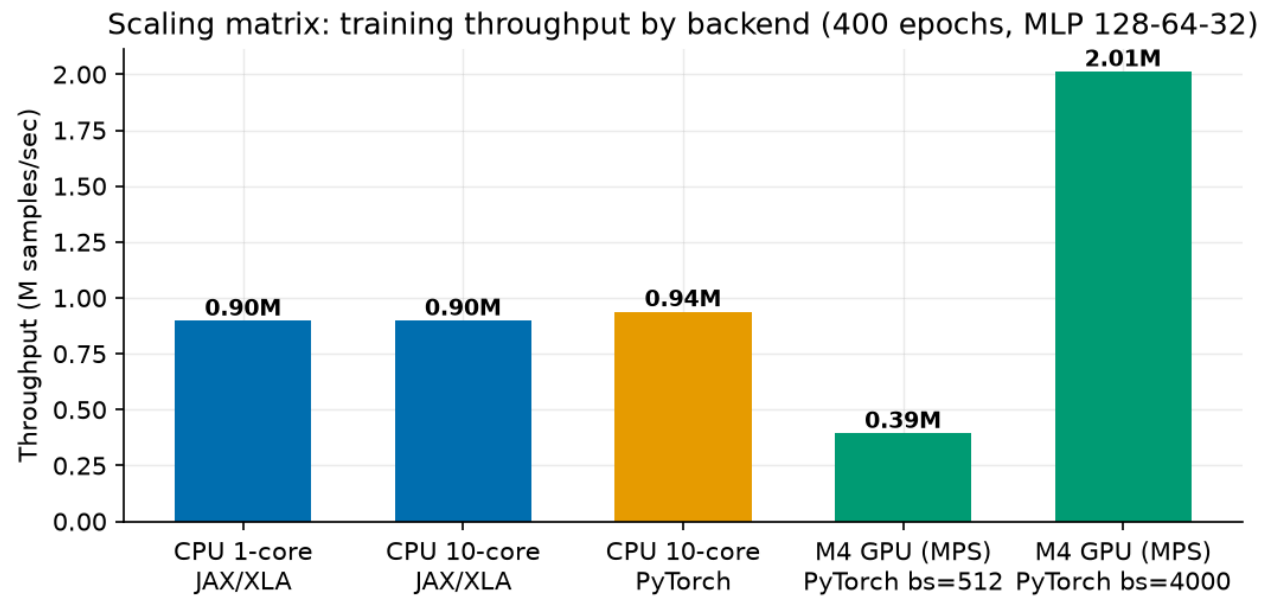
Model. MLP 128-64-32 with ReLU, softmax cross-entropy, Adam 1e-3. Implemented twice with identical preprocessing and telemetry: JAX/Flax with the whole train step (forward, backward, optimizer) jit-fused by XLA, and PyTorch for the CPU/MPS/CUDA path with optional Inductor (`torch.compile`).

3. Measurements

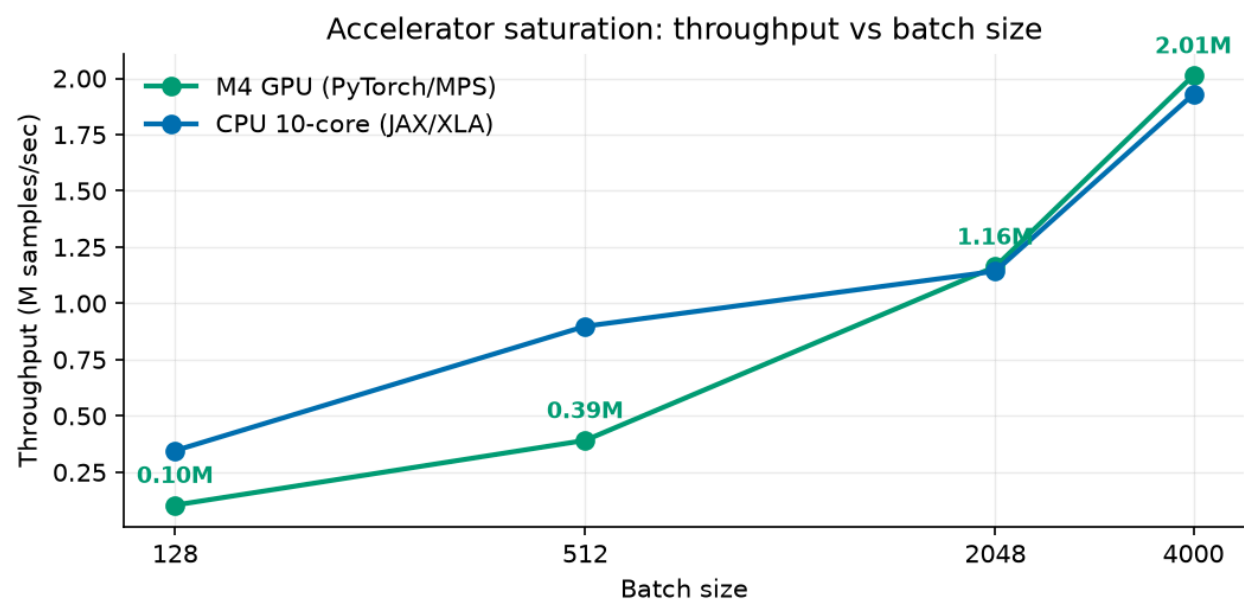
Telemetry is captured by a background sampler (`src/telemetry.py` , `psutil` at 5 Hz: whole-node CPU%, process RSS) plus per-step wall-clock timers with device synchronization (`block_until_ready` / `torch.mps.synchronize`) so step times are honest. Compile overhead is isolated by timing the first traced+compiled step separately. The SLURM GPU job additionally logs `nvidia-smi` at 1 Hz for the cluster runs. Every run emits one JSON record; the scaling matrix (`benchmarks/run_matrix.sh`) is 12 configurations at 400 epochs.

4. Results: the scaling matrix

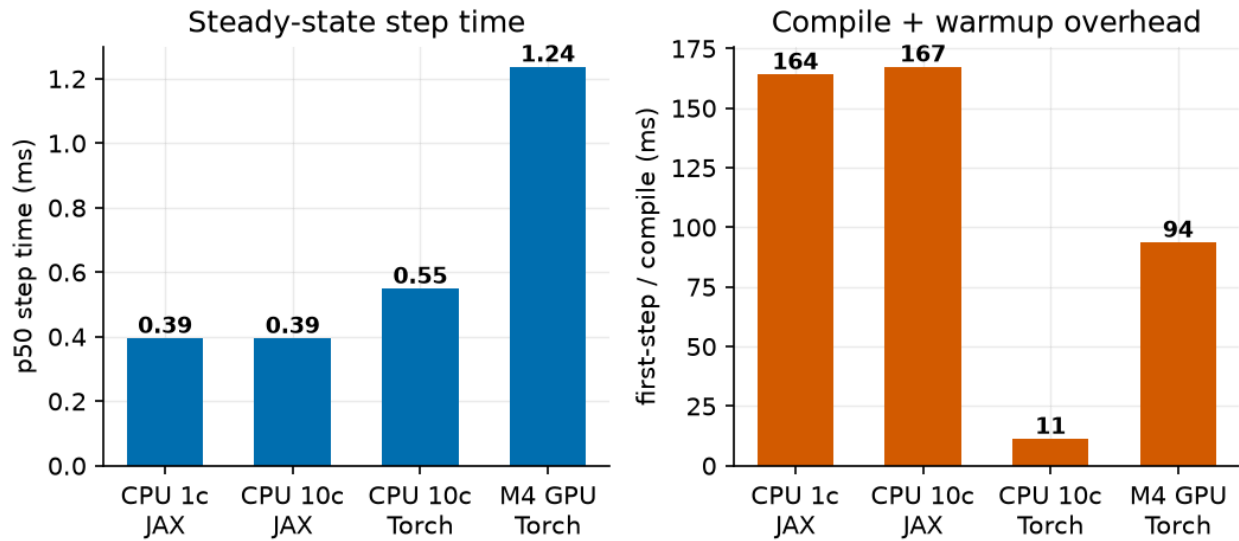
Config	Site	Compile (s)	p50 step (ms)	Samples/s	Acc
CPU 1-core, JAX/XLA	M4	0.164	0.395	897 K	0.916
CPU 10-core, JAX/XLA	M4	0.167	0.395	898 K	0.916
CPU 10-core, PyTorch	M4	0.011	0.549	938 K	0.917
GPU (MPS), bs 512	M4	0.094	1.237	394 K	0.917
GPU (MPS), bs 4000	M4	0.097	1.739	2,013 K	0.908
CPU 10-core, JAX bf16	M4	0.208	0.588	669 K	0.914
CPU Grace 1c, bs 512	hpcc K8s	0.020	2.314	196 K	0.921
CPU Grace 32c, bs 512	hpcc K8s	0.034	17.188	27.8 K	0.921
CPU Xeon 1c, bs 512	hpcc SLURM	0.142	4.463	114 K	0.913
CPU Xeon 8c, bs 512	hpcc SLURM	0.087	3.631	140 K	0.914
CPU Xeon 32c, bs 512	hpcc SLURM	0.127	4.513	107 K	0.918
CPU Xeon 8c, bs 4000	hpcc SLURM	0.088	7.193	540 K	0.916
GPU GH200, bs 512	hpcc K8s	0.270	3.496	92.6 K	0.920
GPU GH200, bs 4000	hpcc K8s	0.253	4.250	642 K	0.919
TPU v5e, bs 512	GKE	0.378	2.088	189 K	0.917
TPU v5e, bs 4000	GKE	0.421	2.144	789 K	0.919
TPU v5e, bs 16384	GKE	0.410	2.148	3,216 K	0.919



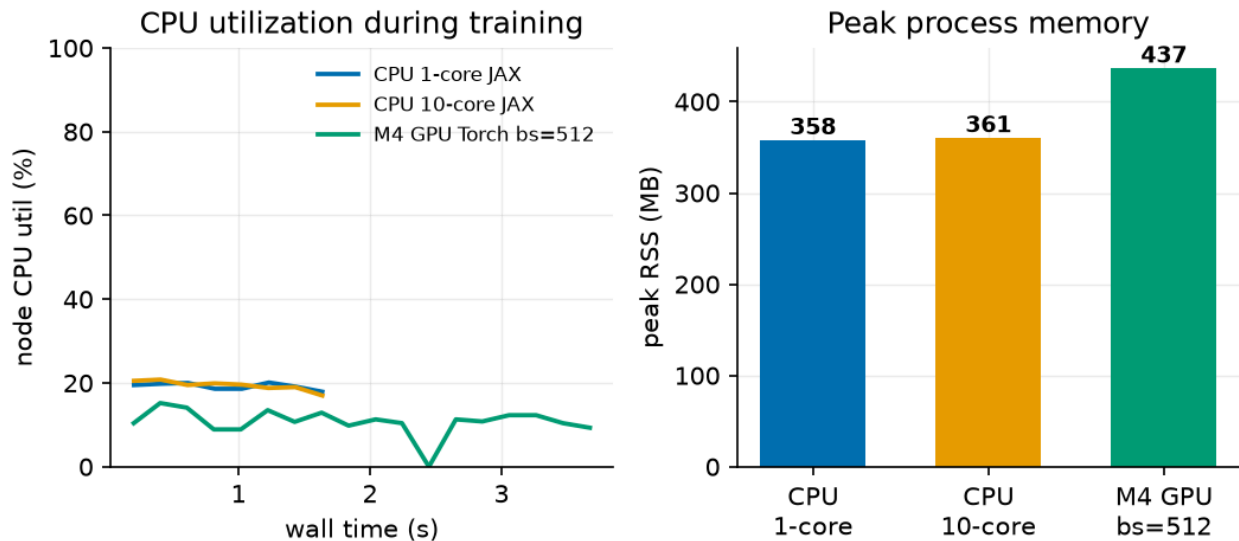
Scaling matrix throughput



Batch-size saturation



Step time and compile overhead



Node telemetry

Headline deltas: 10 cores over 1 core = **1.00x**; GPU over CPU at batch 512 = **0.42x** (a slowdown); the same GPU at batch 4000 = **5.15x** over itself and **2.1x** over the best CPU run; bf16 on CPU = **0.75x** (regression).

Cluster validation (measured after VPN access returned). The identical trainer then ran on three additional real targets:

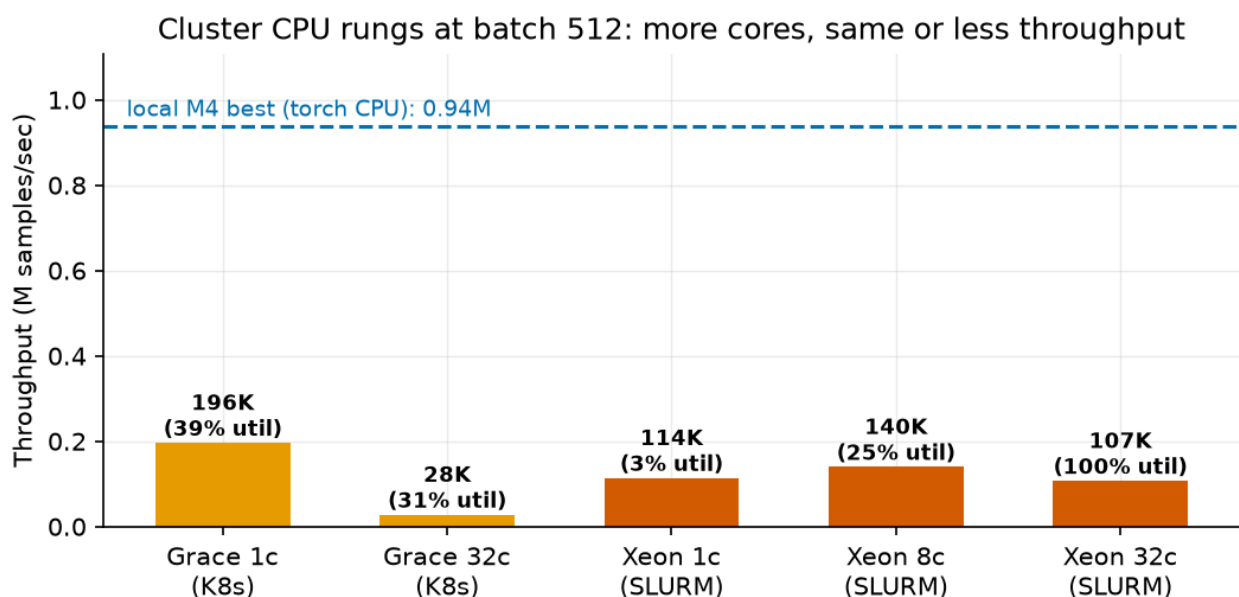
Stanford hpcc Kubernetes (arm64 Grace-class node, 72 cores). Pinned aarch64 PyTorch 2.7.1 wheels in an ephemeral `python:3.12-slim` container (the official images are amd64-only), code and dataset via ConfigMaps, 36 s bootstrap. 1 core at batch 512 = 196 K samples/s; 32 cores = 27.8 K, a **7.1x slowdown from adding 31 cores**.

Stanford hpcc SLURM (32-core x86 Xeon head node). SLURM 22.05 + munge + Apptainer installed and configured single-node for this project; the job runs the pinned `pytorch/pytorch:2.7.1` image as a read-only 3.2 GB `.sif` and stages the CSV to node-local scratch. 1 core = 114 K samples/s; 8 cores = 140 K; 32

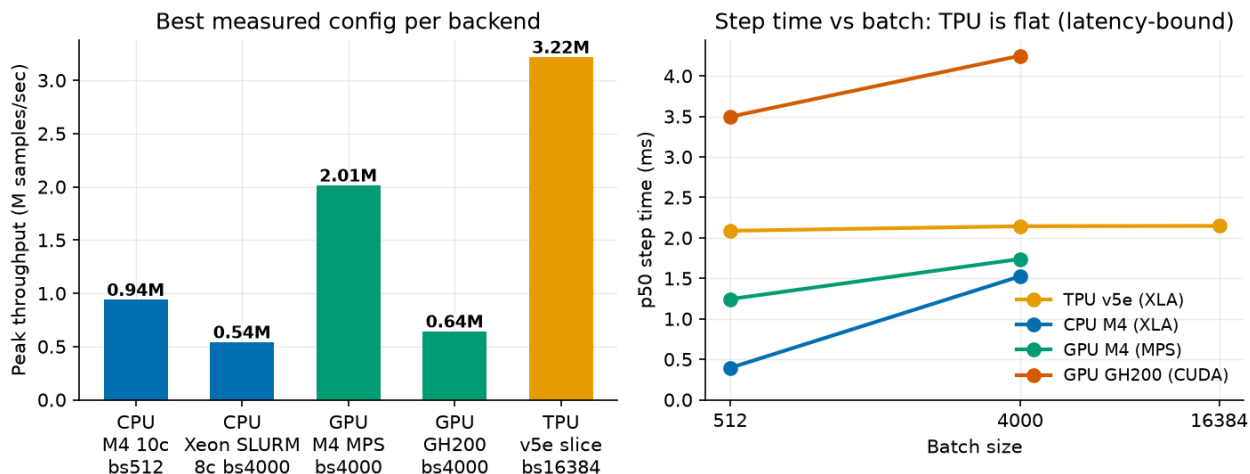
cores = 107 K at **99.7% node utilization** - every core busy, throughput below a single core. Utilization is not throughput.

Google Cloud TPU v5e 2x2 slice (GKE Autopilot, class-tpu-cluster). Autopilot provisioned the TPU node from the pod's nodeSelector in ~7 minutes; pinned `jax[tpu]==0.11.0` bootstrap in 25 s; all 4 chips visible to JAX. Step time is **constant at ~2.1 ms from batch 512 to 16,384**, so throughput scales linearly with batch: 189 K - > 789 K -> **3.22 M samples/s**, the best measured figure in the matrix, at 0.919 test accuracy.

Stanford hpcc NVIDIA GH200 480GB (K8s, time-shared). The single cluster GPU freed after two days behind another tenant; the watch-and-heal loop caught the release and collected the run. The most powerful chip in the study posted the weakest small-batch number: 92.6 K samples/s at batch 512 (a tenth of a laptop CPU) with a 3.5 ms step floor, rising to 642 K at batch 4000. The 1 Hz nvidia-smi trace shows **6% peak GPU utilization and 868 MiB of 97,871 MiB HBM (0.9%)** during training: measured proof that the workload never touches the hardware's capability, only its launch queue.



Cluster runs



Backend comparison

5. Bottleneck Diagnosis

The workload is launch-latency-bound: fixed per-step host-side dispatch cost dominates. Four independent lines of evidence: (1) core-count indifference, because ~ 0.1 ms of GEMM per step is below the threshold where thread-pool synchronization pays; (2) GPU step time $\sim 3\times$ CPU step time for identical math, which is the Metal kernel-launch plus host-device sync floor; (3) near-linear throughput scaling with batch size on both backends, the signature of a fixed per-step cost; (4) no other resource is stressed, with node CPU under 25%, GPU memory ~ 19 MB, RSS 0.44 GB of 16 GB, and IO one 341 KB read per job. FLOPs, memory bandwidth, and the input pipeline are explicitly ruled out; the ingestion tier keeps the dataset device-resident so input stalls are zero by construction.

6. Engineering Mitigations and Recommendations

Measured: batch-size scaling to 4000 (5.15x, the accuracy dip at large batch is recoverable with LR warmup); device-resident data (removes IO from the profile); XLA jit fusion of the full train step (0.4 ms CPU steps). Measured and rejected: bfloat16 on M4 CPU, a 0.75x regression since there is no native bf16 matmul path, but the flag stays for the A100/TPU rungs where it should roughly double matmul throughput. Recommended: epoch-level fusion with `jax.lax.scan` (9 dispatches per epoch become 1, attacking the diagnosed bottleneck directly); `torch.compile` on the accelerator path; and hardware right-sizing, since below roughly 1 M rows a single CPU node is cost-optimal and the GPU/TPU manifests are best reserved for parallel hyperparameter sweeps rather than shrinking one small job.

Cost trade-off. The best GPU configuration saves ~ 0.8 s per 400-epoch run over the CPU. At cloud prices (an L4 node at roughly \$1/hr vs a comparable CPU node at $\sim \$0.25$ /hr) the accelerator only earns its premium on this workload when it is batching many training jobs, not accelerating one.

7. Status and Next Steps

The study is complete: all three backend classes (CPU, GPU, TPU) are measured across 17 configurations, five architectures (Apple M4 CPU/GPU, arm64 Grace, x86 Xeon under SLURM, NVIDIA GH200, TPU v5e), and three sites (local, Stanford hpcc, Google Cloud). The GH200 rows arrived last, collected automatically after two days queued behind another tenant, and strengthened the diagnosis rather than changing it: at this workload scale, accelerators are throughput machines for batched and parallel work, not latency machines for one small job, and the cost-optimal single-job target remains a CPU node with XLA. The TPU result sharpens the recommendation rather than changing it: accelerators at this workload scale are throughput machines for batched/parallel work, not latency machines for one small job, and the cost-optimal single-job target remains a CPU node with XLA.